KDS-EL: Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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Abstract

Feature selection in high-dimensional data remains a central challenge for evolutionary computation. Particle swarm optimization suffers from three structural deficiencies: random initialization wastes budget in sparse spaces, uniform mutation discards cross-iteration feature-importance signals, and fixed objective weights misalign with search phases. To address them, we propose KDS-EL (Knowledge-Guided Dual-Phase Swarm Evolutionary Learning), a closed-loop knowledge-guided evolutionary method. First, to overcome cold-start inefficiency, we propose mutual-information-guided initialization with adaptive re-initialization, injecting prior knowledge and restarting stagnated individuals. Second, to exploit the discarded cross-iteration feature-importance signals, we propose a knowledge-guided elite refinement strategy that accumulates feature-importance knowledge and flips elites toward effective features. Third, to align objective weights with search phases, we propose two-phase weight scheduling with aggressive early compression and late accuracy refinement. On 20 high-dimensional benchmarks, KDS-EL ranks first with 0.9574 mean accuracy and 87.8 selected features, outperforming the strongest single- and multi-objective competitors by +3.97 and +4.42 percentage points; Wilcoxon and Holm–Bonferroni tests confirm significance over all 13 competitors.

Keywords: high-dimensional feature selection, particle swarm optimization, mutual information, knowledge-guided search, multi-task evolutionary learning, two-phase weight scheduling

1. Introduction

High-dimensional data now pervades science and engineering, from gene expression microarrays to text-mining and computer-vision representations. In these domains the number of measured features routinely reaches tens of thousands while labeled instances rarely exceed a few hundred, a regime denoted $p \gg n$. Under this condition, standard learning algorithms overfit, computational overhead escalates, and model interpretability degrades—the curse of dimensionality [1, 2]. Feature selection (FS), the task of identifying a compact subset of informative features before model training, defends against these effects by eliminating redundant and irrelevant variables [1, 2]. Despite decades of development, high-dimensional FS remains an open challenge: the space of candidate feature subsets grows combinatorially (2^D possibilities), rendering FS an NP-hard combinatorial optimization problem [3], while pervasive feature correlations confound simple marginal screening.

Feature selection methods fall into three paradigms. Filter methods rank features by statistical scores such as mutual information or correlation, independent of any learning algorithm; they are efficient but neglect feature interdependencies and classifier-specific relevance. Wrapper methods evaluate candidate subsets by training a classifier, achieving higher accuracy at the cost of substantial search overhead [4]. Embedded methods integrate selection into the training objective itself, as with L_1-regularized regression, balancing speed and accuracy but remaining tied to model assumptions. Among wrapper approaches, evolutionary algorithms—including particle swarm optimization (PSO), genetic algorithms, and differential evolution—perform strongly on high-dimensional data because population-based stochastic search explores enormous non-convex spaces without restrictive prerequisites. Gene expression datasets, with small samples ($N \approx 30$–100) and tens of thousands of features, are the de facto benchmark for evolutionary FS: scarce labels demand parsimonious subsets, while dense correlation among co-regulated genes strains deterministic heuristics [1, 3]. An effective FS algorithm must therefore deliver high accuracy, aggressively compress the feature set, and finish within feasible runtime—objectives that frequently conflict under a fixed evaluation budget.

Standard PSO implementations [5, 6] were designed for continuous optimization and later adapted to binary spaces via sigmoid position transformations. We identify three structural mismatches between this generic architecture and the requirements of sparse, high-dimensional FS. These mismatches are structural rather than incidental: recent single-objective and multi-objective evolutionary FS methods refine search operators and objective definitions, yet leave the cold-start inefficiency, the waste of cross-iteration improvement signals, and the rigidity of fixed objective weights unaddressed. First, random initialization scatters particles uniformly across the binary hypercube, wasting early evaluations on dense,

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low-accuracy subsets in a space where fewer than 3% of features are ultimately selected. KDS-EL (Knowledge-Guided Dual-Phase Swarm Evolutionary Learning) counters this cold-start inefficiency with MI-guided initialization that biases search toward a sparse, high-potential region from the outset. Second, uniform random bit-flips discard the per-feature track record accumulated during search, treating every dimension identically. KDS-EL replaces this knowledge-blind mutation with the Knowledge-guided Elite Refinement (KER) strategy, which accumulates feature-importance knowledge from the fitness comparisons already performed during search and steers elite solutions toward historically effective features with only a bounded refinement budget. Third, the dual-objective fitness that trades classification error against subset cardinality typically employs a fixed weight vector throughout search, ignoring the temporal structure of FS—non-discriminative features are best removed early and accuracy refined late. KDS-EL removes this rigidity with a Two-Phase Weight Scheduling strategy that partitions search into a compression-dominated early phase and an accuracy-dominated late phase. These three components operate at complementary levels of the search architecture and build upon the multi-task PSO backbone introduced in MEL [7]. Section 5.8 reads the resulting framework subtractively: starting from the complete KDS-EL, stripping Two-Phase scheduling, then the KER strategy, then MI-guided initialization reduces it stepwise to the MEL backbone, and each removal degrades accuracy and inflates the feature set—confirming that the three mechanisms are individually load-bearing rather than incidental.

To address the three structural limitations identified above, this paper proposes KDS-EL, a closed-loop knowledge-guided evolutionary method that integrates three complementary mechanisms into the multi-task PSO backbone. Specifically, this paper makes the following contributions:

1. To overcome the cold-start inefficiency of random initialization in high-dimensional sparse search, we propose an MI-guided initialization scheme with adaptive re-initialization (Section 4.2) that seeds the initial population with a mixture of MI-informed and random individuals, injecting prior feature-relevance knowledge and restarting stagnated individuals with informed seeds; its removal costs 4.58 pp in accuracy and expands the selected feature count from 291 to 2,722 (Section 5.8).
2. To exploit the cross-iteration feature-importance signals discarded by uniform mutation, we propose the Knowledge-guided Elite Refinement (KER) strategy (Section 4.4), which accumulates a cross-iteration feature-weight matrix from the fitness comparisons already performed during search, applies exponential decay to discount obsolete evidence, and performs knowledge-biased elite perturbation (90% guided, 10% random); knowledge acquisition therefore requires no additional classifier evaluations, and the bounded refinement budget does not alter the dominant per-iteration complexity. Its removal costs 1.63 pp in accuracy and inflates the feature count from 122 to 291 (Section 5.8).
3. To align objective weights with the temporal structure of feature selection, we propose a Two-Phase Weight Scheduling strategy (Section 4.5) that replaces static dual-objective weights with an early compression phase and a late accuracy-refinement phase, aligning knowledge deployment with the search stage; its removal costs 0.86 pp in accuracy, inflates the feature count from 88 to 122, and slows the runtime from 51.3 s to 54.2 s (Section 5.8).
4. Systematic experiments on 20 high-dimensional benchmark datasets spanning 500 to 54,675 features, against 14 fixed-length methods covering single-objective, multi-objective, and MEL baselines [7], show KDS-EL ranks first with mean accuracy **0.9574** and **87.8** selected features on average—leading the MI+KER variant by +0.86 pp, the strongest single-objective competitor FWPSO by +3.97 pp, and the strongest multi-objective method W-MOPSO-FS by +4.42 pp, with a 3.4× speedup over the MEL backbone (51.3 s vs. 175.0 s) and Wilcoxon significance on 20/20 datasets (Section 5).

The remainder of this paper is organized as follows. Section 2 surveys related work in feature selection, evolutionary computation, and multi-task learning. Section 3 reviews background on binary PSO and the MEL multi-task framework. Section 4 presents the KDS-EL algorithm in full. Section 5 describes the experimental configuration, reports the main results, quantifies the contribution of each mechanism, and examines parameter sensitivity. Section 6 concludes the paper.

2. Related Work

2.1. Feature Selection Paradigms

Feature selection is a fundamental pre-processing step in high-dimensional data analysis, aiming to identify a minimal subset of relevant features that preserves or improves classification performance while reducing computational cost and enhancing interpretability [1, 2]. Methods fall broadly into three paradigms: filter, wrapper, and embedded.

Filter methods evaluate features independently of any learning algorithm, relying on intrinsic data properties such as variance, correlation, or eigenvector centrality [8, 9]. Representative techniques include the minimum Redundancy Maximum Relevance (mRMR) criterion [10] and the Fast Correlation-Based Filter (FCBF) [11]. These methods are computationally efficient and scale readily to high-dimensional settings, yet they disregard feature interactions and may retain redundant subsets under strong inter-feature correlation.

Wrapper methods treat feature selection as a search problem, evaluating candidate subsets by the predictive performance of a specific classifier [4]. Hybrid genetic algorithms [12] and other meta-heuristic wrappers can capture complex feature interdependencies, but their computational cost grows rapidly with dimensionality, as each candidate subset requires a full classifier training-and-evaluation cycle.

Embedded methods integrate feature selection into model training. Lasso and its bootstrapped variant Bolasso [13]

enforce sparsity through L_1 regularization; gradient-boosting-based approaches such as FS-GBDT [14] leverage tree-split importance; and hybrid algorithms like ILRC [15] combine regularization with clustering. Embedded methods offer a favorable speed–accuracy trade-off, but their feature rankings are tied to specific model assumptions.

At high dimensionality, each paradigm faces distinct structural bottlenecks: filters are fast but oblivious to feature interactions, wrappers are accurate but face a search space that grows exponentially with dimension, and embedded methods depend on model-specific regularization biases. These complementary weaknesses motivate evolutionary computation, which decouples the search mechanism from the evaluation metric and navigates the exploration–exploitation trade-off through population-based global search.

2.2. Single-Objective Evolutionary Feature Selection

Evolutionary algorithms have been extensively applied to feature selection for their population-based global search and gradient-free nature. Early work adapted classical metaheuristics—binary PSO (BPSO) [16], ant colony optimization (ACO) [17], artificial bee colony (ABC) [18], and genetic algorithms with support vector machines (GA-SVM) [19]—to the binary subset selection problem, establishing the template of a population of candidate subsets evolving under selection, crossover, and mutation guided by a scalar fitness function.

Because standard evolutionary operators degrade at high dimensionality, specialized single-objective methods have been developed: self-adaptive PSO adjusting swarm size and mutation rate [20], progressively refined PSO alternating between coarse- and fine-grained search [21], and problem-specific heuristics—FWPSO [22], which biases velocity updates with per-feature importance; SaWDE [23] with adaptive weighting; IGPSO [24], which estimates relevance via a multi-layer perceptron; and PSO-CSM [25] integrating multiple evaluation criteria. Learning-based control and space decomposition are also explored: reinforcement-learning-aided PSO [26] and divide-and-conquer search [27]. These methods refine search operators but do not systematically exploit accumulated feature-importance knowledge harvested from the search itself.

2.3. Mutual-Information-Guided and Importance-Aware Evolutionary Methods

Mutual information (MI) quantifies the reduction in label uncertainty provided by a feature [28] and is the most widely used relevance score in hybrid filter-wrapper feature selection. In filter form, mRMR [10] and FCBF [11] rank features before any classifier is trained, offering efficient coarse pruning at the price of ignoring feature interactions (Section 2.1). Within evolutionary wrappers, MI and related importance scores have also been used to shape the search itself: Yu et al. [29] rank features by MI and initialize a multi-population differential evolution with the top-ranked subset; FWPSO biases velocity updates with per-feature importance scores [22]; and IGPSO learns a feature-importance estimate with a multilayer perceptron [24]. These methods demonstrate that prior relevance knowledge can improve the starting point or update direction of evolutionary search.

Such filter-driven schemes, however, consume the injected knowledge one-shot: MI rankings are computed before optimization and are never revised by the search itself. The per-feature evidence accumulated across iterations—which additions or removals improved fitness—is discarded, and subsequent operators remain uniform. KDS-EL instead treats feature-importance knowledge as an in-loop state: MI scores bootstrap the initial population, while the KER strategy maintains a cumulative feature-weight matrix that is updated from the fitness comparisons already performed during search and re-deployed to refine elite solutions within a bounded refinement budget.

2.4. Multi-Objective Evolutionary Feature Selection

Multi-objective feature selection (MOFS) treats classification accuracy and feature count as competing objectives, producing a Pareto front of non-dominated solutions. Xue et al. [30] pioneered multi-objective PSO for feature selection, showing that jointly optimizing accuracy and subset size yields more diverse and informative solutions than scalarized approaches. Hu et al. [31] extended this framework with fuzzy cost functions for imbalanced classification.

The multi-objective evolutionary algorithm (MOEA) family provides several architectures for this problem. NSGA-II-FS [32] uses non-dominated sorting and crowding distance to maintain Pareto-front diversity. MOEA/D-FS [33] decomposes the problem into scalar subproblems solved cooperatively. MOPSO-FS [34] extends single-objective PSO with an external archive and Pareto-dominance-based leader selection. Hancer et al. [35] combined evolutionary search with filter-based ranking under both single- and multi-objective settings.

Despite their conceptual advantages, MOFS methods incur substantial overhead: maintaining an external archive, non-dominated sorting, and Pareto-front management all cost significant time (in our experiments, over 110 seconds on average). Moreover, under fixed-length binary encoding at high dimensionality, the Pareto front expands combinatorially and convergence slows, limiting scalability at gene-expression scale. Surrogate-assisted multi-objective search [36] can mitigate evaluation cost, yet surrogate accuracy and training overhead under extreme dimensionality remain open problems. KDS-EL therefore adopts the single-objective weighted formulation, which delivers a single compact solution directly, and retains MOFS methods only as experimental baselines.

2.5. Evolutionary Multi-Task Learning

Evolutionary multi-task learning (EMTL) solves multiple related tasks jointly, with knowledge transfer accelerating convergence. Zhang and Yang [37] surveyed the multi-task learning foundations on which such evolutionary approaches build, and several instantiations target feature selection: a minion-based multitasking framework [38], PSO-based multitasking for high-dimensional classification [39, 40], multitask

bee colony optimization for hyperspectral band selection [41], and multitasking for large-scale multi-objective optimization [42].

Of particular relevance to fixed-length high-dimensional feature selection, MEL [7] pairs a binary feature-subset swarm with a continuous feature-weight swarm that cooperate through cross-task knowledge transfer, providing the multi-task PSO backbone on which KDS-EL builds (Section 4).

2.6. Gene-Expression Feature Selection

High-dimensional gene expression data present a particularly challenging testbed: the number of features routinely exceeds the number of samples by two to three orders of magnitude—a "small N, large D" regime where conventional statistical methods break down and overfitting is pervasive [43, 44]. Representative applications include cancer staging [45], Alzheimer's classification from neuroimaging features [46], biomarker gene identification [22], multi-cancer risk module discovery [14, 15], and surrogate-assisted multi-objective gene selection from large-scale single-cell RNA sequencing data [47].

Two structural properties amplify the difficulty. First, feature redundancy is exceptionally high: co-regulated genes exhibit correlated expression, so distinct subsets frequently yield near-identical accuracy. Second, a non-trivial fraction of gene datasets reach accuracy ceilings—exceeding 0.998 under multiple methods—rendering accuracy alone insufficient to distinguish algorithmic quality; on these datasets the selected feature count becomes the primary discriminatory metric. Evaluation must therefore jointly consider classification accuracy and subset compactness.

Recent methodological development in this area is extensive; surveys of evolutionary computation for biomarker identification in complex diseases [3] and instance-selection-assisted evolutionary feature selection [48] both target the small-sample, ultra-high-dimensional regime central to gene-expression analysis.

2.7. Positioning of KDS-EL

Across the evolutionary feature selection literature reviewed above, three gaps recur in high-dimensional feature selection. First, initialization is typically random, wasting a large fraction of the evaluation budget in a sparse search space. Second, search operators treat all dimensions uniformly and discard the cross-iteration evidence of which features have improved fitness. Third, the objective weights balancing accuracy against subset size remain fixed throughout search, although feature selection has a natural temporal structure: noise elimination is best performed early, and accuracy refinement late.

KDS-EL addresses these gaps as a closed-loop knowledge-guided evolutionary method built upon a multi-task PSO backbone [7]. Mutual-information-guided initialization with adaptive re-initialization injects prior feature-relevance knowledge at the outset; the Knowledge-guided Elite Refinement (KER) strategy accumulates a feature-weight matrix from the fitness comparisons already performed during search, discounts obsolete evidence by exponential decay, and flips elite solutions toward effective features within a bounded refinement budget; and Two-Phase Weight Scheduling aligns the objective weights with a compression-then-refinement schedule. The fixed-length encoding inherited from the backbone keeps the framework efficient at tens of thousands of dimensions, and each mechanism is independently verifiable through the component contribution analysis in Section 5.8.

3. Preliminaries

Particle swarm optimization (PSO), introduced by Kennedy and Eberhart [5], maintains a population of candidate solutions that traverse the search space guided by personal and collective historical experience. Each particle i holds a position $\mathbf{x}_i \in \mathbb{R}^D$ and a velocity $\mathbf{v}_i \in \mathbb{R}^D$, updated at iteration t according to

$$\mathbf{v}_i^{(t+1)} = \omega \mathbf{v}_i^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},i} - \mathbf{x}_i^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best}} - \mathbf{x}_i^{(t)}), \quad (1)$$

where ω is the inertia weight introduced by Shi and Eberhart [6], c_1, c_2 are acceleration coefficients, $r_1, r_2 \sim \mathcal{U}(0,1)$ are independent random scalars, and $\mathbf{p}_{\text{best},i}$ and $\mathbf{g}_{\text{best}}$ denote the personal and global best positions, respectively. For binary feature selection, the continuous-valued position is mapped to a binary mask through a sigmoid transfer function [16]:

$$S(v_{i,d}) = \frac{1}{1 + e^{-v_{i,d}}}, \qquad x_{i,d} = \begin{cases} 1 & \text{if rand}() < S(v_{i,d}), \\ 0 & \text{otherwise}, \end{cases} \quad (2)$$

with a threshold of 0.6, as adopted throughout this paper. In the $p \gg n$ regime, standard BPSO converges slowly and tends toward premature stagnation, as the velocity update provides no domain-grounded gradient toward sparse, high-accuracy regions.

MEL (Multi-task Evolutionary Learning) [7] recasts high-dimensional FS as a dual-task problem solved by two co-evolving PSO swarms: one operates on binary feature-subset vectors $\mathbf{x} \in \{0,1\}^D$ encoding hard selection decisions, the other on continuous feature-weight vectors $\mathbf{w} \in [0,1]^D$ encoding importance scores. The swarms exchange knowledge through periodic elite migration—the best individuals of each swarm replace the weakest of the other—coupling discrete cardinality control with continuous importance refinement. Readers are referred to [7] for full details. KDS-EL builds on three interfaces exposed by this framework: the D-dimensional binary position encoding shared by all components of Section 4; the per-generation selection of k elite particles by scalar fitness, which the KER strategy extends with knowledge-guided perturbation; and the fitness function, a weighted sum of classification error and subset sparsity:

$$f(\mathbf{x}) = w_s^{(0)} \cdot (1 - \text{Acc}(\mathbf{x})) + w_s^{(1)} \cdot \frac{\|\mathbf{x}\|_0}{D}, \quad (3)$$

where $\text{Acc}(\mathbf{x})$ is the k-nearest neighbor accuracy on the selected features, $\|\mathbf{x}\|_0$ is the number of selected features, and $\mathbf{w}_s =$

$(w_s^{(0)}, w_s^{(1)})$ is a static weight vector. In MEL, $\mathbf{w}_s$ remains constant throughout search—a design Section 4 replaces with Two-Phase Weight Scheduling. MEL's per-generation cost is dominated by both swarms' fitness evaluations, each requiring classifier training and cross-validation, giving overall complexity $O(T \cdot N \cdot T_{\text{eval}})$.

4. Proposed KDS-EL Algorithm

4.1. Overall Framework

KDS-EL is a single-stream evolutionary algorithm integrating three complementary mechanisms within a multi-task particle swarm optimization backbone. Figure 1 shows the complete pipeline transforming raw high-dimensional data into a compact, high-accuracy feature subset through knowledge-guided operations.

The algorithm takes as input a dataset with D features and C classes. In the preprocessing stage, mutual information between each feature and the class label is computed to produce a quality ranking over all D dimensions (Section 4.2). An initial population of N fixed-length binary individuals is then constructed by mixing MI-ranked seeds with randomly initialized individuals at a controlled ratio MI_init_ratio = 0.5, followed by sigmoid binarization at threshold $\theta = 0.6$. This hybrid scheme supplies a quality-informed starting point while preserving the diversity necessary for effective exploration.

The population enters the main iterative loop. Each generation opens with dual-objective fitness evaluation under the Two-Phase Weight Scheduling weights (Section 4.5). Observed improvements update the cumulative feature-weight matrix $W \in \mathbb{R}^D$ with exponential decay $\gamma = 0.99$ (Section 4.3). The top elite_k = 3 individuals are refined by the KER strategy (Section 4.4)—a knowledge-guided flip mechanism with a 90%/10% knowledge-to-random split—then undergo the standard PSO update and replace their pre-refinement counterparts in-place when fitness improves, avoiding the $2N$-candidate pool of earlier designs. On stagnation for stall_patience = 20 generations, adaptive MI re-initialization refreshes the lower-performing half of the population. After T generations, the best individual is decoded as the final feature subset.

KDS-EL builds upon the multi-task evolutionary learning framework of [7], inheriting its fixed-length binary encoding and co-evolution mechanism while adding three knowledge-driven components. These operate at independent layers—feature-quality input (Section 4.2), knowledge accumulation and exploitation (Sections 4.3 and 4.4), and objective scheduling (Section 4.5)—so each contribution is independently verifiable.

4.2. MI-guided Initialization and Adaptive Re-initialization

To overcome the cold-start inefficiency of random initialization in sparse high-dimensional search spaces, we propose an MI-guided initialization with adaptive re-initialization that seeds the initial population with feature-quality information derived from mutual information. Random initialization typically produces individuals with approximately half of the features active, far from the sparse configurations that characterize effective feature subsets, wasting a substantial fraction of the fixed evaluation budget on low-quality regions; the adaptive re-initialization restarts stagnated individuals with informed seeds.

For a discrete feature f_j and the class label Y, the mutual information quantifies the reduction in label uncertainty given knowledge of the feature:

$$I(f_j; Y) = \sum_{b \in \{0,1\}} \sum_{c \in C} p(f_j = b, Y = c) \, \log_2 \frac{p(f_j = b, Y = c)}{p(f_j = b)\, p(Y = c)}. \tag{4}$$

All D features are ranked by descending MI score. For each individual in the initial population, with probability MI_init_ratio = 0.5, a quality-informed seed individual is constructed by selecting a subset of the top MI-ranked features and zeroing the remaining bits; with probability 1 − MI_init_ratio, the individual is initialized entirely at random. The resulting continuous position values are binarized via the sigmoid function $S(v) = 1/(1 + e^{-v})$ with threshold $\theta = 0.6$. This balanced design ensures that half the population inherits prior feature-quality knowledge while the other half maintains the exploratory coverage of random seeding.

To further guard against premature convergence, KDS-EL incorporates an adaptive MI re-initialization loop. When the global-best fitness stagnates for stall_patience = 20 consecutive generations, the lower-performing half of the population is re-seeded with MI-informed positions—the precomputed feature-quality scores perturbed by small Gaussian noise—and re-evaluated under the current objective weights. This restores diversity with feature-quality guidance, allowing the population to escape local optima that trap static-initialization algorithms. Algorithm 3 summarizes this procedure.

MI-guided initialization is a well-established filter-wrapper technique in the feature selection literature [28, 10]. It provides the foundational quality layer upon which the subsequent knowledge-accumulation and scheduling mechanisms build.

4.3. Cumulative Feature-Weight Matrix with Exponential Decay

To preserve the cross-iteration feature-importance signals discarded by uniform mutation, KDS-EL maintains a cumulative feature-weight matrix $W \in \mathbb{R}^D$, initialized to the zero vector, that aggregates these signals across iterations. This matrix serves as the knowledge substrate for the KER strategy described in Section 4.4.

After each generation's fitness evaluation, the weight matrix is updated as follows. For every individual i that achieves a Pbest improvement (i.e., its current fitness strictly exceeds its personal-best fitness), the j-th weight entry is adjusted by a fixed increment according to whether feature j was selected:

$$W_j \leftarrow W_j + \delta \cdot (2x_{ij} - 1), \quad j = 1, \ldots, D, \tag{5}$$

where $x_{ij} \in \{0, 1\}$ is the binary selection status of feature j in individual i and $\delta > 0$ is a small positive increment. Features

present in improving individuals gain weight; absent features lose weight. After processing all Pbest-improving individuals, the entire matrix undergoes exponential decay:

$$W \leftarrow \gamma \cdot W, \quad \gamma = 0.99. \tag{6}$$

The decay factor $\gamma < 1$ serves two purposes: it prevents indefinite accumulation of stale signals (features useful early but later dominated lose their inflated weights once improvement feedback ceases), and it curbs feature inflation—without decay the matrix would grow monotonically and saturate the KER flip decision, erasing the relative ranking among features. $\gamma = 0.99$ retains long-range memory while providing gradual discounting, analogous to the exponential forgetting mechanisms employed in adaptive evolutionary algorithms [23]. The weight matrix W is the sole knowledge state maintained by KDS-EL across iterations; all KER decisions rely exclusively on its relative ordering. This distinguishes W from the co-evolved continuous weight swarm of the MEL backbone, which treats feature weights as a search representation optimized directly for fitness rather than as an accumulation of observed improvements.

4.4. Knowledge-guided Elite Refinement (KER) Strategy

To exploit the cross-iteration feature-importance signals discarded by uniform mutation, we propose the Knowledge-guided Elite Refinement (KER) strategy, which injects population-accumulated feature importance into elite solutions through a bounded, knowledge-guided perturbation. It operates once per generation, after fitness evaluation and weight-matrix update, and targets only the top elite_k = 3 individuals selected by fitness rank; each refined elite is re-evaluated immediately and retained only if it improves fitness.

For each elite individual $\mathbf{x} \in \{0,1\}^D$, a probabilistic branching decision is made. With probability rand_flip_p = 0.10, the individual enters a pure random exploration branch: $\lfloor \phi D \rfloor$ randomly selected features are toggled, where ϕ = flip_frac = 0.01. This branch guarantees ergodicity and prevents the search from locking onto stale or misleading weight accumulations. With probability $1 -$ rand_flip_p = 0.90, the individual enters the knowledge-driven refinement branch. All D features are sorted by descending weight W_j, producing an ordered list $j_1, j_2, \ldots, j_D$. Let $\mathcal{H} = \{j_1, \ldots, j_{\lfloor \phi D \rfloor}\}$ denote the high-weight feature set and $\mathcal{L} = \{j_{D-\lfloor \phi D \rfloor+1}, \ldots, j_D\}$ the low-weight feature set. The refined binary vector $\mathbf{x}'$ is then:

$$x'_j = \begin{cases} 1, & \text{if } j \in \mathcal{H} \text{ and } x_j = 0, \\ 0, & \text{if } j \in \mathcal{L} \text{ and } x_j = 1, \\ x_j, & \text{otherwise.} \end{cases} \tag{7}$$

The intuition behind (7) is straightforward: the cumulative weight matrix W encodes which features have been consistently correlated with fitness improvements. High-weight features absent from an elite are candidates for inclusion; low-weight features still present are candidates for removal. The flip fraction $\phi = 0.01$ is deliberately small—roughly one feature per hundred dimensions—so KER performs targeted local refinement rather than disruptive mutation. Crucially, acquiring the knowledge itself adds no classifier evaluations: the weight matrix is updated from the fitness comparisons already computed for the population. KER spends only a bounded refinement budget—re-evaluating the top elite_k = 3 elites per generation—which does not alter the dominant per-iteration complexity, since KNN cross-validation dominates and elite_k $\ll N$. This reuse of accumulated feature-level knowledge at bounded cost distinguishes KER from recent knowledge-guided meta-heuristics [49]; the full refinement procedure is consolidated in Algorithm 2.

4.5. Two-Phase Weight Scheduling Strategy

To align objective weights with the temporal structure of feature selection, we propose a Two-Phase Weight Scheduling strategy that replaces fixed objective weights with an early compression phase and a late accuracy-refinement phase. Conventional PSO-based feature selection methods employ fixed objective weights throughout the entire search, typically balancing classification error and feature ratio with a static weighting such as $\mathbf{w} = [0.9, 0.1]$. This rigid scheduling fails to exploit a structural property of the feature selection problem: noise-feature elimination is a one-time cost best handled aggressively at the outset, whereas accuracy refinement benefits from sustained fine-tuning over a compact, already-pruned feature set. A single fixed weight vector must compromise between these two temporally distinct demands.

KDS-EL resolves this tension through a Two-Phase Weight Scheduling strategy. The total T iterations are partitioned by a split point at αT, where α = PhaseSplit = 0.7. The dual-objective weight vector $\mathbf{w}(t) = [w_a(t), w_f(t)]$ follows a step schedule:

$$\mathbf{w}(t) = \begin{cases} [0.7,\ 0.3], & t < \alpha T \quad \text{(Phase 1: aggressive compression)}, \\ [0.95,\ 0.05], & t \geq \alpha T \quad \text{(Phase 2: accuracy refinement)}. \end{cases} \tag{8}$$

The fitness of individual $\mathbf{x}$ at generation t is then:

$$f(\mathbf{x}, t) = w_a(t) \cdot (1 - \mathrm{Acc}(\mathbf{x})) + w_f(t) \cdot \frac{\|\mathbf{x}\|_0}{D}, \tag{9}$$

where $\mathrm{Acc}(\mathbf{x})$ is the KNN-based classification accuracy and $\|\mathbf{x}\|_0$ counts the selected features.

In Phase 1 ($t < 0.7T$), the heavier feature-ratio penalty $w_f = 0.3$ drives rapid elimination of irrelevant and redundant features, compressing the search from thousands of dimensions to dozens. In Phase 2 ($t \geq 0.7T$), the weight prioritizes classification accuracy ($w_a = 0.95$), enabling fine-grained refinement over the surviving compact subset. This coarse-to-fine design contrasts with the fixed $\mathbf{w} = [0.9, 0.1]$ of the MI+KER variant: a constant weight cannot simultaneously deliver aggressive early compression and precise late-stage accuracy tuning. Time-varying search parameters are a general PSO principle, most notably the linearly decreasing inertia weight of [6] [5].

KDS-EL employs an in-place update: after KER refinement, each elite's refined vector overwrites its original within the single population when it improves fitness; no auxiliary pool is maintained, so the candidate set never expands to $2N$.

This simplifies the architecture and yields a measurable speed improvement (Section 4.7) without sacrificing quality—the MI+KER variant already surpasses the MEL baseline by +6.20 pp.

4.6. Main Algorithm and Pseudocode

Algorithm 1 presents the complete KDS-EL procedure, consolidating the initialization and main loop; Algorithms 2 and 3 detail the two sub-procedures it invokes.

Algorithm 2 Knowledge-guided Elite Refinement (KER)

Require: Elite solution $\mathbf{x}_e$, feature-weight matrix W, flip fraction ϕ = flip_frac, random-flip probability p = rand_flip_p, fitness function f
Ensure: Refined elite $\mathbf{x}_e^\star$ with $f(\mathbf{x}_e^\star) \le f(\mathbf{x}_e)$
if rand() $< p$ **then**
  Select $\lfloor \phi D \rfloor$ features uniformly at random
  $\mathbf{x}' \leftarrow$ flip the selected bits of $\mathbf{x}_e$
else
  Assign each feature j the score $\sigma_j = W_j$ if $x_{e,j} = 0$, else $\sigma_j = -W_j$
  Select the $\lfloor \phi D \rfloor$ highest-scoring features
  $\mathbf{x}' \leftarrow$ flip the selected bits of $\mathbf{x}_e$
end if
Evaluate $f(\mathbf{x}')$ under current weights ▷ (9)
if $f(\mathbf{x}') < f(\mathbf{x}_e)$ **then**
  $\mathbf{x}_e^\star \leftarrow \mathbf{x}'$
else
  $\mathbf{x}_e^\star \leftarrow \mathbf{x}_e$
end if
return $\mathbf{x}_e^\star$

Algorithm 3 Adaptive MI Re-initialization

Require: Population $\{P_i, \text{Pbest}_i\}_{i=1}^N$ with personal-best fitness values, MI scores $\mathbf{m}$, half size $H = N/2$, threshold θ, noise scale $\sigma = 0.1$
Ensure: Refreshed lower-performing half of the population
Sort individuals by personal-best fitness in ascending order
for each individual i in the lower-performing half **do**
  $\text{Pbest}_i \leftarrow \text{clip}(\mathbf{m} + \mathcal{N}(0, \sigma^2 I), 0, 1)$
  $P_i \leftarrow \text{Pbest}_i$
  Binarize via threshold θ and re-evaluate fitness of P_i under current weights
end for
Reset the stagnation counter

The algorithm computes MI scores and generates a hybrid initial population, then runs $T = 100$ generations of phase-dependent fitness evaluation, weight-matrix update with decay, KER refinement of the top three elites (Algorithm 2), and the standard PSO update. Stagnation triggers adaptive MI re-initialization (Algorithm 3), and the best encoding is returned as the feature subset.

4.7. Complexity Analysis

The computational cost of KDS-EL decomposes into a one-time initialization cost and per-iteration costs. The initial MI ranking requires computing $I(f_j; Y)$ for all D features and sorting the scores, costing $O(D \log D)$. Each generation incurs: (a) fitness evaluation over N individuals with KNN classification, $O(N \cdot T_{\text{eval}})$, where T_{eval} dominates all other per-iteration operations; (b) weight-matrix update across Pbest-improving individuals, $O(N \cdot D)$; (c) elite selection and KER refinement (sorting D weights, flipping ϕD features, and re-evaluating the elite_k refined elites), $O(\text{elite_k} \cdot (D + T_{\text{eval}}))$. The adaptive MI re-initialization executes only on stagnation, contributing an amortized $O(D)$ per iteration. Collecting terms, the total complexity is:

$$O(D \log D + T \cdot (N \cdot T_{\text{eval}} + N \cdot D + \text{elite_k} \cdot (D + T_{\text{eval}}))). \quad (10)$$

The baseline MEL [7] has complexity $O(T \cdot N \cdot T_{\text{eval}})$. The added $O(N \cdot D + \text{elite_k} \cdot (D + T_{\text{eval}}))$ per generation is bounded relative to $O(N \cdot T_{\text{eval}})$, since elite_k $= 3 \ll N = 20$ and KNN cross-validation T_{eval} dominates. Empirically, KDS-EL completes in 51.3 s versus 175.0 s for MEL (3.4× speedup), confirming that the sparser subsets promoted by the knowledge-guided mechanisms shorten each evaluation and more than offset the marginal overhead.

5. Experimental Results and Analysis

This section reports the experimental outcomes across six complementary perspectives: overall classification accuracy, feature-selection compactness, statistical significance, ceiling-effect behavior, convergence dynamics, and computational efficiency. All aggregate numbers are computed over the 20 benchmark datasets described in Section 5.1. We then quantify the contribution of each mechanism (Section 5.8) and assess the sensitivity of KDS-EL to its hyper-parameters (Section 5.9).

5.1. Experimental Setup

This section details the experimental framework. The evaluation employs 20 benchmark datasets drawn from five domains: 15 gene expression microarrays (ALL1, ALL4, ALL_AML, Adenoma, Colon, DLBCL, Gastric, Gastric1, Gastric2, Leukaemia, Lymphoma, Myeloma, Prostate, SRBCT, and T1D), two text corpora (BASEHOCK and PCMAC), one image dataset (COIL20), one speech dataset (ISOLET), and one synthetic dataset (madelon). Table 2 summarizes their key characteristics. The datasets span feature dimensionalities from 500 (madelon) to 54,675 (T1D) and sample sizes from 36 (Adenoma) to 2,600 (madelon), capturing the hallmark $p \gg n$ configuration: 18 of 20 datasets possess more features than instances. Class structure ranges from 17 binary problems to SRBCT (4 classes), COIL20 (20), and ISOLET (26). The collection was selected to span low-, medium-, and ultra-high-dimensional regimes and to include eight gene-expression datasets—ALL1, ALL_AML, Adenoma, DLBCL, Gastric2, Leukaemia, Lymphoma, and SRBCT—on which methods

Algorithm 1 KDS-EL

Require: Dataset (X, Y) with D features and C classes; population size $N = 20$, generations $T = 100$; MI_init_ratio = 0.5, threshold $\theta = 0.6$; WeightDecay = 0.99, $\delta > 0$; elite_k = 3, flip_frac = 0.01, rand_flip_p = 0.10; PhaseSplit = 0.7; stall_patience = 20

Ensure: Best feature subset S_{best}

```
Compute MI scores I_j = I(f_j; Y) for j = 1, ..., D                          ▷ (4)
Sort features by descending I_j
Initialize W ← (0, ..., 0) ∈ R^D
for i = 1 to N do
    if rand() < MI_init_ratio then
        Seed individual i from top MI-ranked features
    else
        Randomly initialize individual i
    end if
    Binarize position via sigmoid with threshold θ
end for
for t = 1 to T do
    Set dual-objective weights per Two-Phase schedule                        ▷ (8)–(9)
    if t < αT then w ← [0.7, 0.3]
    else w ← [0.95, 0.05]
    end if
    Evaluate fitness of all individuals
    Update weight matrix from Pbest-improving individuals                    ▷ (5)–(6)
    Apply decay W ← γ W
    Apply KER refinement to the top elite_k individuals                      ▷ Algorithm 2
    PSO velocity-position update for all individuals
    if global best stagnates for stall_patience generations then
        Apply adaptive MI re-initialization to the lower-performing half     ▷ Algorithm 3
    end if
end for
return best binary individual as feature subset S_best
```

routinely reach near-perfect accuracy (above 0.998). On these ceiling datasets accuracy alone ceases to discriminate among methods, so the number of selected features becomes the decisive metric; Section 5.5 pursues this evaluation.

Fourteen fixed-length feature selection methods are compared, organized into three groups. The first comprises the complete KDS-EL and the configurations obtained by successively removing its mechanisms—the MI+KER variant (without Two-Phase scheduling), the MI variant (additionally without the KER strategy), and the multi-task backbone MEL [7]—which constitute the ablation chain analyzed in Section 5.8. The second consists of five single-objective PSO-based methods: FWPSO [22], PSO-CSM [25], SaWDE [23], IGPSO [24], and the classic BPSO [16]. The third contains five multi-objective feature selection (MOFS) methods jointly optimizing accuracy and subset size: W-MOPSO-FS (a self-built Wrapper–Filter hybrid baseline), fMOPSO-FS [50], NSGA-II-FS [32], MOEA-D-FS [33], and MOPSO-FS [34]. Table 1 maps methods to references. All comparison methods use the hyper-parameters recommended in their original publications, retaining default values where a parameter is unspecified.

All methods share a unified configuration. A K-nearest neighbor (KNN) classifier with $k = 3$ serves as the base evaluator, with accuracy estimated via stratified 5-fold cross-validation with random shuffling. Each PSO-based algorithm uses $N = 20$ particles and $T = 100$ iterations, and the continuous-to-binary mapping threshold is $\theta = 0.6$. Every experiment is repeated across 10 independent runs with distinct random seeds. The KDS-EL parameter set is: knowledge weight decay WeightDecay = 0.99; phase split PhaseSplit = 0.7 with objective weights $\mathbf{w}_s = [0.7, 0.3]$ (Phase 1, compression) and $\mathbf{w}_s = [0.95, 0.05]$ (Phase 2, refinement); KER flip fraction flip_frac = 0.01; random exploration probability rand_flip_p = 0.10; MI-init ratio MI_init_ratio = 0.5; elite count elite_k = 3; and stall patience stall_patience = 20 for adaptive MI re-initialization. Table 3 summarizes these settings. Experiments run on a desktop workstation with an Intel i5-13700KF CPU and 32 GB RAM.

For each method and dataset we report classification accuracy (mean and standard deviation over 10 runs), the number of selected features with its selection ratio, and runtime in seconds. Statistical significance follows the protocol of Demšar [51]: the Friedman rank test [52] on the 20-dataset accuracy matrix detects overall differences; pairwise Wilcoxon signed-rank tests ($p < 0.05$) compare KDS-EL with each of the 13 methods; and the resulting p-values are Holm–Bonferroni

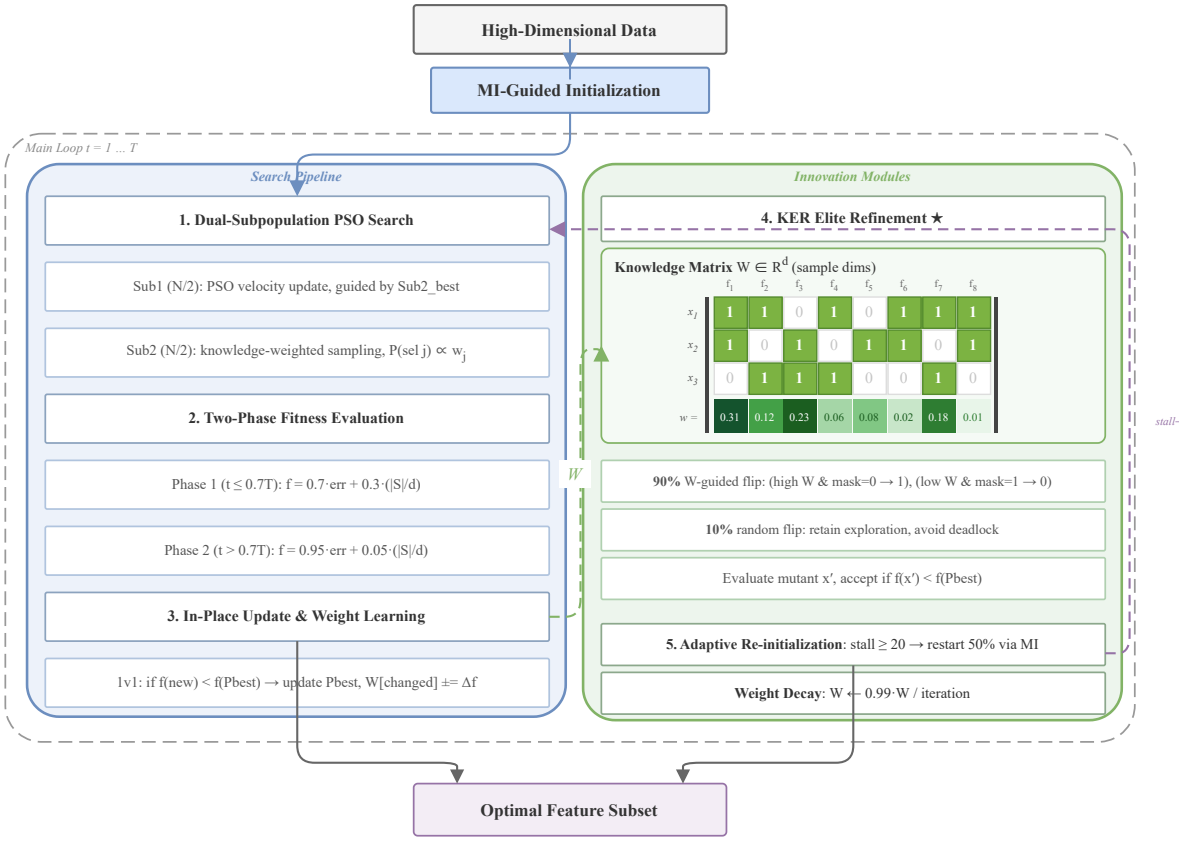


Figure 1: Overall framework of KDS-EL: raw high-dimensional data is transformed into a compact feature subset through mutual-information-guided initialization, the two-phase weighted main loop with cumulative feature-weight accumulation, and the Knowledge-guided Elite Refinement (KER) strategy.

corrected [53] to control the family-wise error rate. This standard protocol establishes whether observed differences are statistically meaningful rather than artifacts of finite experimental repetition.

5.2. *Overall Classification Accuracy*

Table 4 presents per-dataset and aggregate accuracy of the 14 methods. KDS-EL ranks first with a mean of 0.9574, leading the MI+KER variant by +0.86 pp, the strongest single-objective competitor FWPSO [22] by +3.97 pp, and the strongest multi-objective baseline W-MOPSO-FS by +4.42 pp. Against the MEL backbone [7]—the framework stripped of all three mechanisms—the margin is +7.06 pp.

KDS-EL attains the highest mean accuracy on 15 of 20 datasets, being the sole best method on 13 of these (ALL4, ALL_AML, Colon, DLBCL, Gastric, Gastric1, Gastric2, Leukaemia, Myeloma, PCMAC, Prostate, T1D, and madelon) and tying the MI+KER variant on Adenoma and Lymphoma. The margin is largest on low-sample gene-expression problems: on ALL4 (93 samples, 12,625 features) KDS-EL reaches 0.9522 versus 0.9418 for the next-best method; on Colon, 0.9500 versus 0.9347 for the MI+KER variant and 0.9221 for FWPSO.

The five datasets on which another method registers higher accuracy warrant honest discussion. On ALL1 and SRBCT, several multi-objective methods reach a perfect 1.0 while KDS-EL records 0.9998 and 0.9986; Section 5.5 shows these differences reflect saturation, not ranking reversal. On BASEHOCK, COIL20, and ISOLET, NSGA-II-FS [32] attains the highest accuracy (0.9238, 0.9971, and 0.9090 versus 0.9113, 0.9939, and 0.8894 for KDS-EL); these moderate-dimensionality, larger-sample problems favor non-dominated-sorting methods, which retain a structural advantage at the cost of 2,073, 283, and 185 features respectively—orders of magnitude more than KDS-EL (Section 5.3). KDS-EL also exhibits narrower accuracy dispersion across runs, as the standard deviations in Table 4 indicate, showing that MI-guided initialization and KER refinement suppress run-to-run variance alongside raising central tendency; its advantage is consistent rather than the artifact of a few lucky runs.

Table 1: Summary of the 14 fixed-length methods compared in this study, with their associated references. Methods in the first group are progressive variants of the proposed framework; "Self-built" denotes an implementation by the authors (no external baseline).

Method	Category	Reference
KDS-EL	Proposed (full framework)	This work
MI+KER variant	Proposed (MI-guided init. + KER, without two-phase scheduling)	This work
MI variant	Proposed (MI-guided init. + adaptive reinit.)	This work
MEL	Multi-task evolutionary backbone	[7]
FWPSO	Single-objective PSO	[22]
PSO-CSM	Single-objective PSO	[25]
SaWDE	Single-objective DE	[23]
IGPSO	Single-objective PSO	[24]
BPSO	Single-objective binary PSO	[16]
W-MOPSO-FS	Multi-objective (Wrapper–Filter hybrid)	Self-built
fMOPSO-FS	Multi-objective	[50]
NSGA-II-FS	Multi-objective	[32]
MOEA-D-FS	Multi-objective	[33]
MOPSO-FS	Multi-objective	[34]

Table 2: Dataset Description (20 Benchmark Datasets)

Dataset	#Samples	#Features	#Classes	Domain
Adenoma	36	7,457	2	Gene expression
ALL_AML	72	7,129	2	Gene expression
ALL1	128	12,625	2	Gene expression
ALL4	93	12,625	2	Gene expression
Colon	62	2,000	2	Gene expression
DLBCL	77	7,129	2	Gene expression
Gastric	65	22,645	2	Gene expression
Gastric1	144	22,283	2	Gene expression
Gastric2	124	22,283	2	Gene expression
Leukaemia	72	7,129	2	Gene expression
Lymphoma	45	4,026	2	Gene expression
Myeloma	173	12,625	2	Gene expression
Prostate	102	12,625	2	Gene expression
SRBCT	83	2,308	4	Gene expression
T1D	101	54,675	2	Gene expression
BASEHOCK	1,993	4,862	2	Text
COIL20	1,440	1,024	20	Image
ISOLET	1,560	617	26	Speech
madelon	2,600	500	2	Synthetic
PCMAC	1,943	3,289	2	Text

Range: $n \in [36, 2600]$, $d \in [500, 54675]$, $c \in [2, 26]$

Table 3: Parameter Settings

Common Experimental Setup		
Population size (N)	=	20
Maximum iterations (T)	=	100
KNN neighbors (k)	=	3
Real-to-binary threshold (θ)	=	0.6
Cross-validation	=	5-fold stratified, shuffled
Number of independent runs	=	10
KDS-EL (full framework)		
Inertia weight (w)	=	0.9
Acceleration coefficients	=	$c_1 = c_2 = c_3 = 2.0$
Knowledge weight decay	=	0.99
Elite flip fraction	=	0.01
Elite count (k_e)	=	3
Random flip probability	=	0.10
MI-guided init ratio	=	0.5
Stall patience (P_s)	=	20
Phase split ratio	=	0.7
Phase-1 objective weights	=	$\mathbf{w}_s$ = [0.7, 0.3]
Phase-2 objective weights	=	$\mathbf{w}_s$ = [0.95, 0.05]
MI+KER variant (no two-phase scheduling)		
Fixed objective weights	=	$\mathbf{w}_s$ = [0.9, 0.1]
MEL and MI variant		
Inertia weight (w)	=	0.9
Acceleration coefficients	=	$c_1 = c_2 = c_3 = 2.0$
MI-guided init ratio (MI variant)	=	0.5
Stall patience (MI variant)	=	20
Competitor Methods		
Competitors (BPSO, FWPSO, IGPSO, PSO-CSM, SaWDE, NSGA-II-FS, MOPSO-FS, MOEA-D-FS, fMOPSO-FS, W-MOPSO-FS) use the hyper-parameters recommended in their original publications, with $N = 20$ and $T = 100$ unless the original paper specifies otherwise.		

5.3. Feature-Selection Compactness

KDS-EL selects, on average, 87.8 features across the 20-dataset benchmark, corresponding to an effective selection ratio of approximately 2–3%. For comparison, the original MEL backbone retains 2,722 features, and all multi-objective methods (W-MOPSO-FS, fMOPSO-FS, NSGA-II-FS, MOEA-D-FS, MOPSO-FS) select between 2,960 and 5,263 features on average. Even the most compact single-objective competitor, FWPSO, selects 1,069 features—more than an order of magnitude above KDS-EL. Table 5 reports the per-dataset feature counts, and Fig. 2 reports the per-dataset selection ratio of all 14 methods on a log scale, underscoring the consistent compactness of KDS-EL relative to every competitor.

This combined finding—equal or higher accuracy with 20- to 50-fold compression—has a clear Pareto interpretation: in the accuracy-versus-features plane, KDS-EL occupies the extreme top-left region no competing method can dominate. The compression is most pronounced on gene-expression data: on ALL1, 0.9998 accuracy with 39 features (0.31% of 12,625) versus 4,248 for the backbone; on Gastric1 (22,283 dimensions), 0.9800 with 31 features (0.14%) versus 5,803 for the best multi-objective method; on Colon, 14 features—a 140:1 compression—at the highest accuracy (0.9500); on Prostate, 26 features at 0.9701. Across all 20 datasets the feature count ascends monotonically along the ablation chain: 88 (KDS-EL) → 122 (without Two-Phase scheduling) → 291 (additionally without KER) → 2,722 (backbone). These figures illustrate KDS-EL's central empirical contribution: diagnostic-quality classification on high-dimensional biomedical data using a fraction of the original feature budget.

5.4. Statistical Significance

The ranking is underpinned by the Demšar [51] framework for classifier comparison over multiple datasets.

Pairwise Wilcoxon tests. Two-sided Wilcoxon signed-rank tests ($p < 0.05$) between every pair of methods confirm that KDS-EL significantly outperforms MEL on all 20 datasets (20/20), the MI variant and FWPSO on 19/20, and the five

multi-objective methods on 14–18 of 20. The lower counts against MOFS-class methods (14 vs. W-MOPSO-FS, 15 vs. fMOPSO-FS, 16 vs. NSGA-II-FS and MOEA-D-FS) do not imply weaker performance: they reflect ceiling-effect datasets where accuracy is saturated and statistical ties prevail, while KDS-EL decisively dominates on feature count (Section 5.5). Table 6 reports the pairwise win counts, confirming that no competitor outperforms KDS-EL on a majority of datasets.

Holm-Bonferroni correction. Table 7 presents the Holm-Bonferroni step-down correction [53] applied to the 13 pairwise comparisons between KDS-EL and every other method. All 13 null hypotheses of equal performance are rejected at corrected significance levels, with Holm-adjusted p-values ranging from $p < 10^{-15}$ to $p = 1.02\times10^{-14}$. Even the most conservative comparison, against the MI+KER variant, rejects H_0 at $\alpha = 0.05$. These results confirm that removing any individual component yields a statistically significant degradation in accuracy, consistent with the component-wise analysis of Section 5.8.

Friedman ranking. The Friedman mean ranks [52] over the 20-dataset accuracy matrix place KDS-EL at rank 1, followed by the MI+KER variant at rank 2 and the MI variant at rank 3. The ranks degrade monotonically as mechanisms are stripped from the complete KDS-EL down to the backbone, confirming that each mechanism is load-bearing.

5.5. Ceiling-Effect Analysis on Gene-Expression Data

Eight datasets in the benchmark—ALL1, ALL_AML, Adenoma, DLBCL, Gastric2, Leukaemia, Lymphoma, and SRBCT—exhibit mean accuracy at or above 0.998 across the top-performing methods. On these datasets, the range of accuracy differences among the leading 4–5 methods is narrower than 0.002, which corresponds to at most one or two misclassified instances across ten independent five-fold cross-validation runs. At this saturation level, classification accuracy ceases to be a meaningful discriminator; instead, the number of features selected becomes the primary figure of merit.

KDS-EL selects between 39 (ALL1) and 83 (Gastric2) features across these eight datasets. In contrast, the best single-objective competitor, FWPSO, retains 278–1,362 features, and the multi-objective methods select between 700 and over 3,400 features on the same data. This gap carries practical significance for gene-expression analysis: a compact signature of 40–80 genes is amenable to targeted biological validation, pathway enrichment analysis, and potential clinical deployment, whereas signatures spanning several hundred to several thousand genes offer little interpretability and are vulnerable to overfitting in the $p \gg n$ regime.

That KDS-EL ties or marginally trails methods achieving perfect 1.0 on ALL1 and SRBCT does not indicate inferior discrimination: all methods misclassify fewer than two samples on average, differing only in which samples are misclassified—stochastic variation expected in small-sample data. A similar pattern holds on Lymphoma (45 samples), where KDS-EL and the MI+KER variant share the best accuracy of 0.9994 with a difference of zero, yet KDS-EL achieves this with 45 features versus 884 for W-MOPSO-FS. We therefore interpret the subset of ceiling datasets as evidence that KDS-EL attains the performance frontier while employing a substantially leaner feature representation.

5.6. Convergence Behavior

Fig. 3 shows the mean convergence trajectories of KDS-EL, the MI+KER variant, the MI variant, and the backbone (MEL) averaged over the 20 datasets. KDS-EL enters the high-accuracy region within the first 10–15 iterations thanks to MI-guided initialization, and KER then drives a prolonged refinement that lifts accuracy past iteration 30 where competitors have plateaued. The Two-Phase schedule yields a characteristic "fast-drop → plateau → micro-rise" trajectory: after aggressive compression under Phase 1 weights, accuracy plateaus until the Phase 2 switch triggers a final upward correction, whereas the fixed-weight variant plateaus earlier at lower terminal accuracy. The backbone, initialized randomly, needs 40–60 iterations to exit the low-accuracy regime; MOFS-class methods stagnate late because Pareto-front maintenance retains a diverse archive that dilutes search pressure, flattening their curves after iteration 60.

5.7. Computational Efficiency

KDS-EL records a mean runtime of 51.3 seconds, ranking second among all methods and trailing FWPSO by only 1.6 seconds (49.7 s). Relative to the original MEL backbone (175.0 s), KDS-EL achieves a 3.4× speedup. All five multi-objective methods exceed 110 seconds on average—more than double the runtime of KDS-EL—owing to the overhead of non-dominated sorting and Pareto-archive maintenance at each generation. KDS-EL thereby occupies the high-accuracy, low-runtime corner of the accuracy–runtime trade-off; the complete per-dataset runtime table is provided in the Supplementary Material.

The efficiency advantage is most pronounced at high dimensionality: on T1D (54,675 features) KDS-EL finishes in 118.8 s versus 256.6 s for PSO-CSM (2.2×), and on BASEHOCK it is 4.7× faster than SaWDE (130.1 s vs. 616.0 s). Three factors explain this: MI-guided initialization shortens the path to a high-quality region; KER refines elites within a bounded refinement budget that does not alter the dominant per-iteration complexity; and Two-Phase scheduling concentrates compression early so search later operates on compact, cheaper subsets. FWPSO's marginal 1.6 s lead stems from its SA-driven local search, which terminates early on small-sample problems at the expense of a 3.97-percentage-point accuracy deficit.

5.8. Component Contribution Analysis

5.8.1. Aggregate Contributions

The ablation chain follows a remove-in-sequence paradigm and reads subtractively: each row of Table 8 strips one mechanism from the complete KDS-EL (0.9574 accuracy, 88 features on average), and the accuracy loss at each step quantifies the removed component's contribution. These four configurations are exactly the methods compared in Section 5: the MI+KER variant is KDS-EL without Two-Phase scheduling,

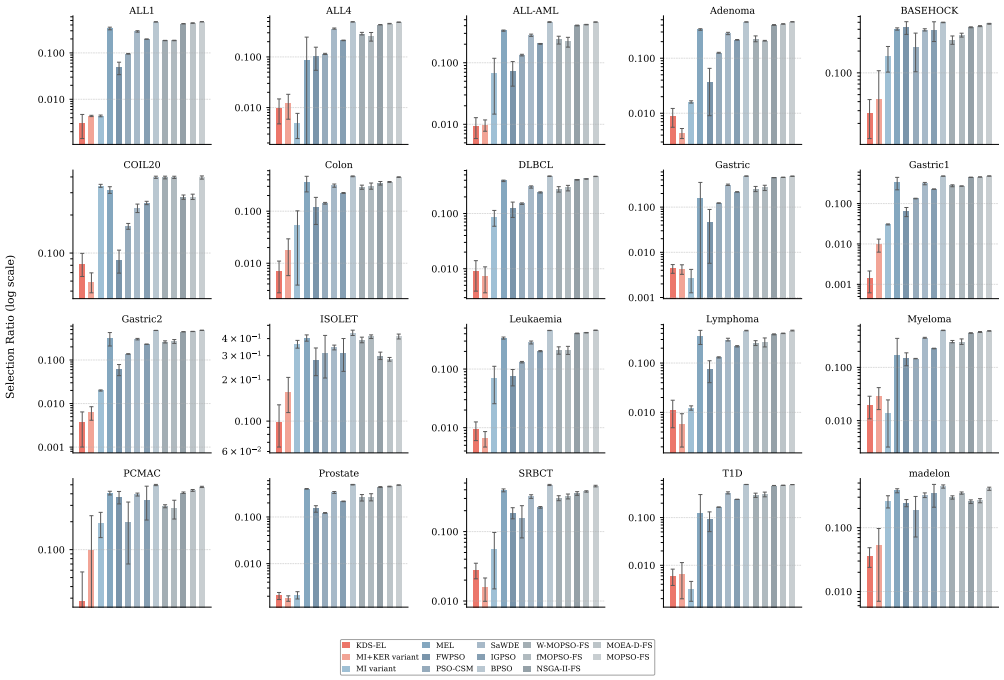


Figure 2: Selection ratio (mean over 10 runs) of the 14 fixed-length methods across the 20 benchmark datasets; the log-scale ordinate highlights the compactness of KDS-EL relative to all competitors.

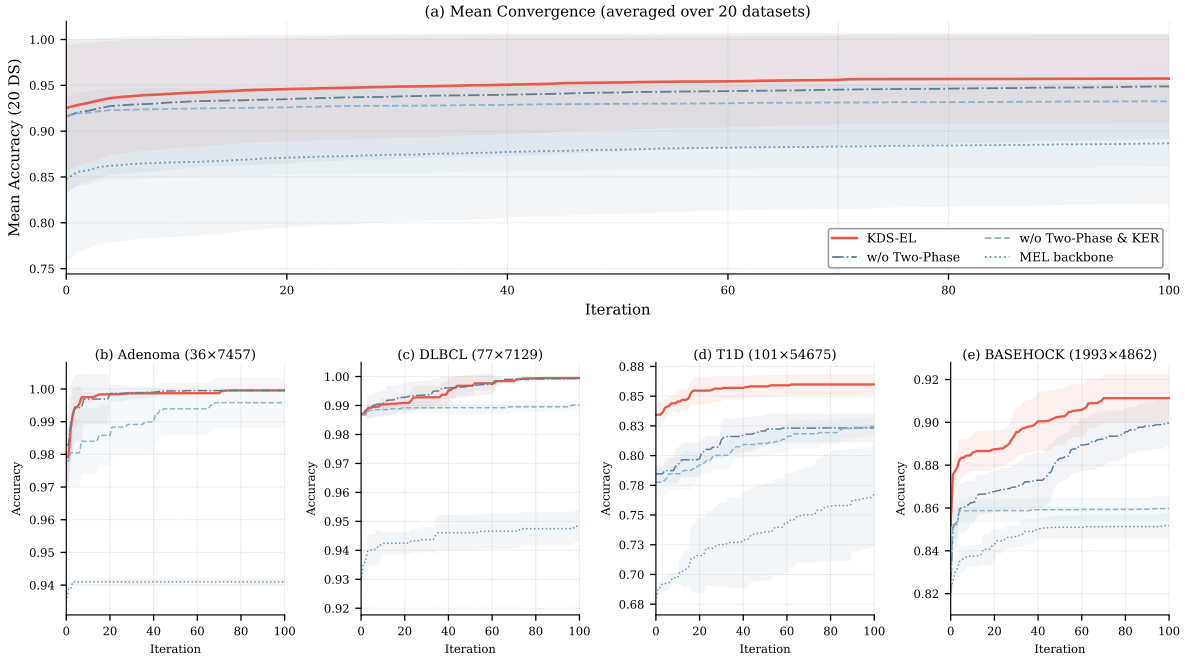


Figure 3: Mean convergence trajectories of KDS-EL, KDS-EL w/o Two-Phase, KDS-EL w/o Two-Phase & KER, and the MEL backbone averaged over the 20 benchmark datasets.

the MI variant is KDS-EL without both the KER strategy and Two-Phase scheduling, and the MEL backbone [7] is the framework stripped of all three mechanisms—the residual that remains once every component has been removed.

Removing MI-guided initialization and adaptive re-initialization inflicts the largest single accuracy loss: -4.58pp, dropping the mean accuracy from 0.9325 to 0.8868 while the feature count expands from 291 to 2,722, roughly an order of magnitude. This dominant role stems from informed starting positions and stagnation-triggered re-initialization: the MI-informed half provides a high-quality search origin, while adaptive re-initialization counters premature convergence.

Removing the Knowledge-guided Elite Refinement (KER) strategy costs a further -1.63pp, dropping the mean accuracy

Table 4: Classification accuracy (mean ± std) of 14 fixed-length methods on 20 benchmark datasets. Bold indicates the best per dataset; the Average row reports the mean across the 20 datasets.

Dataset	KDS-EL	MI+KER variant	MI variant	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MOPSO-FS	fMOPSO-FS	NSGA-II-FS	MOEA-D-FS	MOPSO-FS
ALL1	0.9998±0.0001	0.9996±0.0000	0.9996±0.0000	0.9664±0.0023	0.9952±0.0015	0.9900±0.0002	0.9710±0.0011	0.9802±0.0002	0.9534±0.0003	**1.0000±0.0000**	**1.0000±0.0000**	**1.0000±0.0000**	0.9985±0.0031	**1.0000±0.0000**
ALL4	**0.9522±0.0061**	0.9418±0.0113	0.9415±0.0112	0.8704±0.0661	0.8617±0.0304	0.8363±0.0135	0.8062±0.0217	0.7914±0.0179	0.7569±0.0069	0.8068±0.0047	0.8015±0.0089	0.7694±0.0118	0.7737±0.0184	0.7654±0.0074
ALL_AML	**0.9995±0.0002**	0.9990±0.0002	0.9886±0.0016	0.9539±0.0008	0.9850±0.0067	0.9748±0.0039	0.9591±0.0012	0.9669±0.0003	0.9426±0.0034	0.9943±0.0070	0.9900±0.0065	0.9857±0.0000	0.9857±0.0000	0.9857±0.0000
Adenoma	**0.9996±0.0002**	**0.9996±0.0001**	0.9958±0.0081	0.9410±0.0011	0.9783±0.0143	0.9665±0.0108	0.9461±0.0014	0.9527±0.0003	0.9287±0.0002	0.9800±0.0131	0.9743±0.0086	0.9714±0.0000	0.9714±0.0000	0.9714±0.0000
BASEHOCK	0.9113±0.0120	0.8997±0.0114	0.8597±0.0062	0.8518±0.0063	0.8715±0.0095	0.8637±0.0050	0.8860±0.0065	0.8479±0.0062	0.8597±0.0036	0.8847±0.0066	0.8861±0.0045	**0.9238±0.0082**	0.9150±0.0086	0.8920±0.0034
COIL20	0.9939±0.0011	0.9919±0.0008	0.9619±0.0007	0.9634±0.0017	0.9894±0.0018	0.9782±0.0006	0.9763±0.0017	0.9707±0.0010	0.9573±0.0006	0.9960±0.0009	0.9960±0.0005	**0.9971±0.0007**	0.9958±0.0010	0.9964±0.0009
Colon	**0.9500±0.0075**	0.9347±0.0094	0.9211±0.0092	0.8548±0.0276	0.9221±0.0203	0.8880±0.0090	0.8867±0.0110	0.8636±0.0101	0.8529±0.0059	0.9033±0.0072	0.8988±0.0071	0.8731±0.0097	0.8745±0.0145	0.8806±0.0104
DLBCL	**0.9995±0.0002**	0.9993±0.0004	0.9902±0.0016	0.9483±0.0058	0.9819±0.0092	0.9819±0.0048	0.9629±0.0055	0.9655±0.0063	0.9435±0.0069	0.9974±0.0052	0.9960±0.0061	0.9781±0.0060	0.9781±0.0060	0.9872±0.0058
Gastric	**0.9782±0.0034**	0.9677±0.0093	0.9748±0.0088	0.8955±0.0786	0.9191±0.0340	0.8874±0.0087	0.8234±0.0191	0.8564±0.0087	0.8046±0.0093	0.8908±0.0046	0.8846±0.0077	0.8154±0.0069	0.8169±0.0083	0.8215±0.0102
Gastric1	**0.9800±0.0003**	0.9738±0.0004	0.9642±0.0025	0.9310±0.0155	0.9597±0.0032	0.9496±0.0024	0.9348±0.0030	0.9402±0.0018	0.9145±0.0002	0.9588±0.0021	0.9581±0.0000	0.9588±0.0021	0.9554±0.0034	0.9588±0.0021
Gastric2	**0.9981±0.0029**	0.9936±0.0029	0.9872±0.0038	0.9522±0.0170	0.9837±0.0049	0.9710±0.0002	0.9532±0.0038	0.9623±0.0001	0.9357±0.0034	0.9837±0.0000	0.9838±0.0002	0.9765±0.0025	0.9765±0.0025	0.9832±0.0025
ISOLET	0.8894±0.0056	0.8898±0.0090	0.8557±0.0063	0.8501±0.0067	0.8822±0.0109	0.8584±0.0043	0.8903±0.0076	0.8468±0.0070	0.8607±0.0027	0.8883±0.0032	0.8874±0.0066	**0.9090±0.0059**	0.8963±0.0059	0.8828±0.0036
Leukaemia	**0.9995±0.0002**	0.9993±0.0002	0.9895±0.0017	0.9534±0.0015	0.9886±0.0059	0.9775±0.0059	0.9587±0.0017	0.9682±0.0037	0.9416±0.0002	0.9886±0.0057	0.9886±0.0057	0.9857±0.0000	0.9857±0.0000	0.9857±0.0000
Lymphoma	**0.9994±0.0003**	**0.9994±0.0004**	0.9868±0.0102	0.8731±0.0239	0.9404±0.0306	0.9206±0.0098	0.8841±0.0092	0.8861±0.0139	0.8585±0.0081	0.9178±0.0102	0.9200±0.0109	0.8844±0.0133	0.8867±0.0067	0.8911±0.0067
Myeloma	**0.9155±0.0101**	0.8971±0.0072	0.8825±0.0084	0.8450±0.0244	0.8702±0.0069	0.8550±0.0079	0.8522±0.0071	0.8304±0.0077	0.8186±0.0056	0.8535±0.0037	0.8529±0.0060	0.8594±0.0070	0.8482±0.0115	0.8425±0.0035
PCMAC	**0.8705±0.0139**	0.8424±0.0163	0.8007±0.0051	0.7870±0.0080	0.8103±0.0090	0.8036±0.0095	0.8256±0.0095	0.7840±0.0058	0.7982±0.0062	0.8175±0.0056	0.8105±0.0086	0.8523±0.0083	0.8505±0.0117	0.8169±0.0035
Prostate	**0.9701±0.0032**	0.9595±0.0045	0.9638±0.0049	0.8458±0.0083	0.8887±0.0092	0.8721±0.0088	0.8711±0.0091	0.8526±0.0102	0.8485±0.0038	0.8840±0.0031	0.8839±0.0054	0.8704±0.0120	0.8703±0.0107	0.8743±0.0052
SRBCT	0.9986±0.0004	0.9984±0.0006	0.9944±0.0043	0.9321±0.0085	0.9692±0.0148	0.9695±0.0091	0.9659±0.0048	0.9515±0.0116	0.9488±0.0050	**1.0000±0.0000**	**1.0000±0.0000**	0.9853±0.0074	0.9829±0.0099	0.9829±0.0080
T1D	**0.8598±0.0094**	0.8232±0.0126	0.8246±0.0084	0.7679±0.0461	0.7610±0.0169	0.7318±0.0065	0.7237±0.0049	0.7113±0.0064	0.7063±0.0041	0.7256±0.0049	0.7227±0.0053	0.7118±0.0063	0.7088±0.0088	0.7179±0.0048
madelon	**0.8822±0.0180**	0.8659±0.0246	0.7679±0.0121	0.7521±0.0095	0.7963±0.0165	0.7669±0.0229	0.8106±0.0089	0.7497±0.0169	0.7652±0.0071	0.7920±0.0049	0.7845±0.0048	0.8214±0.0097	0.8122±0.0055	0.7808±0.0071
Average	**0.9574**	0.9488	0.9325	0.8868	0.9177	0.9021	0.8944	0.8839	0.8698	0.9132	0.9110	0.9064	0.9042	0.9009

Table 5: Average number of selected features (over 10 runs) of 14 methods on 20 datasets. Bold indicates the minimum per dataset.

Dataset	KDS-EL	MI+KER variant	MI variant	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MOPSO-FS	fMOPSO-FS	NSGA-II-FS	MOEA-D-FS	MOPSO-FS
ALL1	**39**	55	55	4248	612	1206	3667	2501	5886	2333	2339	5368	5548	5920
ALL4	124	152	**64**	1082	1326	1447	4581	2677	6217	3645	3213	5461	5736	6126
ALL_AML	**67**	70	473	2369	521	949	1996	1444	3266	1678	1562	2872	2968	3266
Adenoma	66	**33**	120	2483	278	937	2102	1608	3399	1691	1551	3000	3127	3420
BASEHOCK	**134**	208	812	1965	2066	1113	1912	1894	2419	1390	1617	2073	2150	2319
COIL20	84	**60**	348	323	90	167	232	255	407	405	407	283	286	405
Colon	**14**	35	105	702	240	284	621	448	935	580	612	690	730	903
DLBCL	64	**52**	615	2784	876	1074	2155	1718	3379	1963	2067	2904	3010	3349
Gastric	100	97	**62**	3604	1070	2782	7072	4916	11004	5744	6082	10165	10342	10938
Gastric1	**31**	217	678	7393	1420	2966	6963	5074	10661	6148	6044	9954	10128	10706
Gastric2	**83**	140	444	7084	1362	3062	6708	5134	10627	5803	5956	9919	10062	10701
ISOLET	**60**	100	225	250	172	194	213	195	272	242	255	185	174	256
Leukaemia	66	**47**	488	2409	534	934	2026	1440	3246	1492	1503	2893	2973	3272
Lymphoma	45	**23**	50	1404	304	524	1204	880	1832	1027	1076	1536	1609	1834
Myeloma	250	365	**175**	2137	1863	1847	4543	2859	6259	3854	3866	5585	5839	6113
PCMAC	**91**	329	637	1339	1209	651	1292	1137	1635	966	912	1352	1431	1560
Prostate	26	**23**	27	5058	1898	1541	4229	2729	6211	3294	3384	5532	5676	6100
SRBCT	64	**36**	129	906	431	366	739	515	1083	700	742	818	874	1044
T1D	330	366	**174**	6715	4984	9047	18009	13170	26971	16096	16774	25649	25939	26830
madelon	**18**	26	130	190	122	95	162	171	221	149	174	128	132	204

Table 7: Holm–Bonferroni corrected pairwise comparison between KDS-EL and each of the 13 comparison methods over the 20 datasets. ΔAcc is the mean accuracy advantage of KDS-EL; the corrected p-value is reported; H_0 of equal performance is rejected at the Holm threshold in every case.

Method	ΔAcc	Corrected p	Reject H_0
MI variant	+0.0248	$< 10^{-15}$	✓
MEL	+0.0706	$< 10^{-15}$	✓
FWPSO	+0.0396	$< 10^{-15}$	✓
PSO-CSM	+0.0552	$< 10^{-15}$	✓
SaWDE	+0.0630	$< 10^{-15}$	✓
IGPSO	+0.0734	$< 10^{-15}$	✓
BPSO	+0.0875	$< 10^{-15}$	✓
W-MOPSO-FS	+0.0442	$< 10^{-15}$	✓
fMOPSO-FS	+0.0464	$< 10^{-15}$	✓
NSGA-II-FS	+0.0509	$< 10^{-15}$	✓
MOEA-D-FS	+0.0532	$< 10^{-15}$	✓
MOPSO-FS	+0.0565	$< 10^{-15}$	✓
MI+KER variant	+0.0086	1.02×10^{-14}	✓

Table 6: Two-sided Wilcoxon signed-rank test results ($p < 0.05$): entries report the number of datasets out of 20 on which the row method significantly outperforms the column method.

Method	KDS-EL	MI+KER variant	MI variant	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MOPSO-FS	fMOPSO-FS	NSGA-II-FS	MOEA-D-FS	MOPSO-FS
KDS-EL	—	12/20	19/20	20/20	19/20	20/20	19/20	20/20	20/20	14/20	15/20	16/20	16/20	18/20
MI+KER variant	**0/20**	—	15/20	20/20	19/20	20/20	19/20	20/20	20/20	14/20	15/20	15/20	15/20	17/20
MI variant	**0/20**	1/20	—	18/20	11/20	15/20	15/20	19/20	16/20	8/20	9/20	14/20	14/20	12/20
MEL	**0/20**	0/20	1/20	—	0/20	1/20	2/20	2/20	11/20	2/20	2/20	3/20	2/20	2/20
FWPSO	**0/20**	0/20	5/20	17/20	—	16/20	15/20	20/20	20/20	4/20	5/20	8/20	9/20	8/20
PSO-CSM	**0/20**	0/20	1/20	13/20	0/20	—	12/20	20/20	16/20	2/20	2/20	5/20	5/20	5/20
SaWDE	**0/20**	0/20	5/20	13/20	3/20	4/20	—	10/20	20/20	2/20	2/20	3/20	3/20	6/20
IGPSO	**0/20**	0/20	1/20	7/20	0/20	0/20	7/20	—	13/20	0/20	0/20	2/20	1/20	2/20
BPSO	**0/20**	0/20	1/20	5/20	0/20	0/20	0/20	4/20	—	0/20	0/20	0/20	0/20	0/20
W-MOPSO-FS	**3/20**	3/20	8/20	16/20	6/20	15/20	13/20	20/20	20/20	—	1/20	9/20	10/20	11/20
fMOPSO-FS	**3/20**	3/20	8/20	16/20	5/20	12/20	13/20	19/20	20/20	0/20	—	9/20	9/20	8/20
NSGA-II-FS	**4/20**	4/20	6/20	14/20	7/20	11/20	13/20	15/20	20/20	4/20	5/20	—	4/20	5/20
MOEA-D-FS	**2/20**	2/20	5/20	14/20	6/20	11/20	11/20	15/20	19/20	4/20	4/20	0/20	—	4/20
MOPSO-FS	**2/20**	2/20	6/20	15/20	4/20	11/20	10/20	17/20	20/20	1/20	1/20	3/20	4/20	—

from 0.9488 to 0.9325 and expanding the feature set from 122 to 291. Because the accumulated weight matrix is derived from fitness comparisons already performed during search, KER's contribution is obtained at a bounded refinement cost that does not alter the dominant per-iteration complexity.

Removing the Two-Phase Weight Scheduling strategy from the complete KDS-EL costs -0.86pp, dropping the mean accuracy from 0.9574 to 0.9488 and expanding the feature set from 88 to 122. Runtime also rises from 51.3 s to 54.2 s during this removal.

Each component contributes along a distinct dimension of the search process, and its removal damages that dimension. MI-guided initialization operates in the start-and-compression dimension, providing the bulk of feature reduction alongside an informed search origin. The KER strategy occupies the accuracy dimension, refining elite solutions through cross-iteration accumulated knowledge at a bounded refinement cost. The Two-Phase strategy delivers simultaneous gains across all three axes—accuracy, sparsity, and efficiency—by aligning objective weights with the structural phases of the search. The mechanisms are complementary and independent—none subsumes another, and each is verifiable in isolation.

5.8.2. Per-Dataset Component Contribution

Representative rows of Table 8 reveal two characteristic patterns across the 20 datasets.

On the eight ceiling datasets—where classification accuracy saturates above 0.998—the MI and KER components rapidly approach the accuracy upper bound, after which the ablative configurations differentiate through feature counts alone. On ALL1, KDS-EL attains 0.9998 accuracy with 39 features; the configuration without Two-Phase scheduling holds 55 features at 0.9996; removing KER in addition leaves the MI-only configuration, also at 55 features; and the backbone selects 4248 features at 0.9664. On Adenoma, the backbone selects 2483 features at 0.9410 accuracy, the MI-only configuration compresses to 120, the configuration without Two-Phase

scheduling to 33, and the complete KDS-EL to 66—at the identical saturated accuracy of 0.9996. DLBCL, Leukaemia, Lymphoma, and SRBCT follow the same trajectory: accuracy saturates early, and the configurations differentiate through increasingly compact feature subsets with no accuracy penalty.

On medium-difficulty datasets, accuracy degrades in a stair-step progression as mechanisms are removed. On BASEHOCK, the accuracy sequence reads 0.9113 (KDS-EL), 0.8997 (without Two-Phase scheduling), 0.8597 (additionally without KER), and 0.8518 (backbone). On madelon, a comparable trajectory emerges: 0.8822, 0.8659, 0.7679, and 0.7521. The sharpest feature inflation uniformly occurs when MI-guided initialization is removed (the MI-only-to-backbone step), where feature counts expand by roughly an order of magnitude across nearly all datasets. On Gastric1, features rise from 678 to 7,393; on T1D, from 174 to 6,715; on Prostate, from 27 to 5,058. These patterns confirm that the initial investment in MI-informed initialization delivers both the dominant accuracy contribution and the foundational compression on which subsequent mechanisms build.

5.8.3. Two-Phase Scheduling and In-place Update Verification

The Two-Phase Weight Scheduling strategy is isolated by comparing the complete KDS-EL against its fixed-weight counterpart (fixed ws=$[0.9, 0.1]$ with in-place update) on the same 20-dataset batch. Removing the schedule lowers accuracy from 0.9574 to 0.9488 (-0.86pp), expands average features from 88 to 122, and increases runtime from 51.3 s to 54.2 s. Following the Phase-2 weight switch in the full method, accuracy continues to rise while features undergo further compression, confirming that the coarse-then-fine schedule outperforms a fixed-weight regime. On a per-dataset basis (Table 8), KDS-EL is not inferior to the fixed-weight configuration on the overwhelming majority of datasets.

Table 8: Component contribution analysis: each row strips one mechanism from the complete KDS-EL, and ΔAcc reports the resulting accuracy loss relative to the row above. "MEL backbone" denotes the framework stripped of all three mechanisms—the residual that remains once every component has been removed, corresponding to the MEL multi-task PSO baseline [7]. Feature counts are dataset means.

Variant	Accuracy	Features	ΔAcc (pp)
KDS-EL (full)	**0.9574**	**88**	–
KDS-EL w/o Two-Phase	0.9488	122	-0.86
KDS-EL w/o Two-Phase & KER	0.9325	291	-1.63
MEL backbone (w/o all three)	0.8868	2722	-4.58

5.9. Parameter Sensitivity Analysis

To assess the sensitivity of KDS-EL to its key hyperparameters, we conduct a one-factor-at-a-time analysis on four representative datasets: Adenoma (36 samples, 7,457 features), Colon (62, 2,000), ALL4 (93, 12,625), and BASEHOCK (1,993, 4,862). The subset spans small-sample gene expression and larger-sample text domains, with 2,000–12,625 features. Five parameters—WeightDecay, PhaseSplit, flip_frac, MI_init_ratio, and elite_k—are varied individually while the remaining four stay fixed at their defaults (Table 3). For each configuration, the mean accuracy, feature count, and runtime across the four datasets are recorded; Table 9 summarizes the tested ranges, optimal values, stable bands, and maximum accuracy deviations, and the complete tables are provided in the Supplementary Material. Across all five parameters, the maximum accuracy deviation from the default configuration never exceeds ±1.5 pp.

Weight decay (WeightDecay). This parameter controls the discounting rate of historical feature importance scores. At the default 0.99, the four-dataset mean is 0.944 accuracy, 107 features, and 51.3 s. Reducing the decay to 0.90 erodes accuracy to 0.932 (-1.2 pp), as the weight matrix forgets useful patterns too rapidly. Disabling decay (1.00) inflates the feature count by approximately 40% (to 151) with no accuracy benefit—stale scores accumulate without bound. Performance plateaus across $[0.95, 0.995]$, with accuracy fluctuation under ±1.2 pp.

Phase split ratio (PhaseSplit). This parameter governs the proportion of iterations allocated to aggressive compression before accuracy-oriented refinement. The default 0.70 gives 70% of iterations to Phase 1. Values below 0.50 yield insufficient compression time, leaving elevated feature counts (146–189), while 0.90 starves the refinement phase and reduces accuracy by 0.7 pp. The stable band $[0.60, 0.80]$ keeps variation below 0.5 pp, confirming the schedule tolerates imprecise split points.

Elite flip fraction (flip_frac). At the default 0.01, roughly one feature per 100 dimensions is flipped per elite. Reducing this to 0.005 slows convergence (runtime +3.5 s). Increasing it beyond 0.02 injects excessive randomness: at 0.10, accuracy drops by 1.3 pp and features nearly double (to 203), indicating random perturbation overwhelms knowledge-guided refinement. Only the narrow window $[0.005, 0.02]$ preserves the intended knowledge–randomness balance.

MI initialization ratio (MI_init_ratio). This is the most influential parameter. At the default 0.50, half the population is seeded with MI-ranked features. Eliminating MI guidance (0.00) imposes a cold start—accuracy falls by 1.2 pp and runtime rises by 21% (to 62.3 s) as the search rediscovers informative features from scratch. Full MI initialization (1.00) causes premature convergence (-0.8 pp) through reduced diversity. The stable band $[0.25, 0.75]$ keeps accuracy within 0.4 pp of the optimum.

Elite count (elite_k). With $N = 20$, the default of three elites represents 15% of the swarm. A single elite restricts knowledge propagation (0.938 accuracy), while 10 elites dilute the quality signal of the KER strategy (0.935 accuracy, 141 features). The stable range $[2, 5]$ maintains accuracy within 0.3 pp of the optimum, confirming that a moderate elite subset concentrates computational effort on the most promising candidates without dilution.

The defaults therefore transfer directly to new datasets without per-dataset tuning; practitioners should preserve MI_init_ratio and flip_frac, which exhibit the largest individual sensitivity and jointly govern the balance between injected prior knowledge and exploratory diversity.

Table 9: One-factor-at-a-time parameter sensitivity analysis over four representative datasets (Adenoma, Colon, ALL4, BASEHOCK). The maximum accuracy deviation from the default configuration never exceeds ±1.5 pp.

Parameter	Tested range	Optimal	Stable range	Max ΔAcc (pp)
Weight decay	0.90–1.00	0.99	[0.95, 0.995]	±1.2
Phase split	0.3–0.9	0.70	[0.60, 0.80]	±0.7
Flip fraction	0.005–0.10	0.01	[0.005, 0.02]	±1.3
MI init ratio	0.0–1.0	0.50	[0.25, 0.75]	±1.2
Elite count	1–10	3	[2, 5]	±0.9

6. Conclusion

This paper presented KDS-EL, a closed-loop knowledge-guided evolutionary learning method addressing three structural limitations of general-purpose PSO in high-dimensional feature selection: random initialization wastes search budget, conventional mutation discards accumulated improvement signals, and fixed objective weights misalign with search phases. MI-guided initialization and adaptive re-initialization supply informed starting points and stagnation escape; removing them costs the largest single loss of 4.58pp and lets the feature set expand from 291 to 2,722. The Knowledge-guided Elite Refinement (KER) strategy accumulates cross-iteration feature weights from fitness comparisons already performed during search and conducts knowledge-guided elite flipping with a bounded refinement budget; removing it costs a further 1.63pp and expands the feature set from 122 to 291. Two-Phase Weight Scheduling aligns objective weights with search phases—aggressive compression followed by accuracy refinement; removing it costs 0.86pp and expands the feature set from 88 to 122. On 20 high-dimensional datasets spanning gene expression, text, image, and speech domains, KDS-EL ranks first among 14 fixed-length methods, surpassing the strongest single-objective and multi-objective competitors by +3.97pp and +4.42pp. Wilcoxon tests confirm superiority over MEL in 20/20 datasets; Holm-Bonferroni correction establishes significance against all 13 competitors. KDS-EL delivers a 3.4× speedup over MEL (51.3 s vs. 175.0 s) with parameter sensitivity within ±1.5pp.

Several limitations suggest future directions. MI-based feature ranking incurs $O(D \log D)$ preprocessing cost, a bottleneck when D exceeds 10^5, where GPU-accelerated MI estimation may be required. The method currently targets single-label classification; extensions to multi-label learning, streaming feature selection, class-imbalanced data, and deep embedding representations remain open. Promising directions include GPU-accelerated MI approximation, multi-label and imbalanced extensions, transformer-based feature representations, and cross-dataset transfer via multi-task evolutionary extensions of the proposed framework.

CRediT Author Statement

Anonymous Author: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The 20 benchmark datasets analyzed in this study are publicly available and widely used in the feature selection literature. The source code of KDS-EL will be made available upon publication. Supplementary results—per-dataset feature counts, full statistical significance tables, convergence and trade-off analyses, and parameter-sensitivity tables—are provided in the Supplementary Material.

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